Multi-Source Web Scraper & Trust Scoring System

1. Scraping Strategy

The project implements a modular, multi-source web scraping architecture designed for high resilience and data integrity across diverse platforms.

Blogs: The scraper uses a multi-tiered approach. It first attempts extraction via newspaper3k for high-quality metadata and body text. If content is insufficient or hidden behind dynamic layouts, a BeautifulSoup fallback is triggered — cleaning the HTML DOM by stripping scripts, styles, and navigation bars, then targeting semantic tags like article, p, and h1 to recover core content. For Medium and TowardsDataScience, specialized heuristics identify specific CSS classes to bypass container noise.

YouTube: Extraction is handled via yt-dlp for metadata and youtube-transcript-api for content. The system prioritizes official transcripts. If transcripts are blocked or unavailable, it falls back to the video description. Authenticated cookies.txt support is included to bypass age restrictions and sign-in requirements.

PubMed: Instead of fragile HTML parsing, the system queries the NCBI Entrez API using BioPython. This extracts metadata, including complex author lists, journal titles, and abstracts, as structured XML, significantly reducing parsing errors common with medical databases.

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2. Topic Tagging Method

A multi-layered NLP pipeline ensures 3 to 5 relevant topic tags are generated for every source, regardless of content length or complexity.

- **RAKE:** The primary engine, identifying key phrases based on word frequency and co-occurrence patterns.
- **SpaCy Noun Chunking:** Extracts meaningful noun phrases to improve the human-readability of generated tags.
- **TF-IDF Fallback:** If fewer than three tags are returned, a TF-IDF model identifies uniquely significant terms against a background corpus.
- **Sanitization:** All tags are post-processed, long phrases over three words are truncated and boilerplate terms such as "subscribe" or "link in bio" are filtered out.

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3. Trust Score Algorithm

The Trust Score is a deterministic heuristic ranging from 0.0 to 1.0, calculated as a weighted combination of five reliability factors:

Trust Score = 0.25 × Author Credibility + 0.20 × Citation Score + 0.20 × Domain Authority + 0.20 × Recency + 0.15 × Medical Disclaimer

- **Author Credibility (25%):** PubMed authors and those from .edu or .gov domains score near 0.9+. Unknown authors or those from unverified domains start at 0.2–0.4.
- **Citation Score (20%):** Source-specific counting, blogs count anchor tags in raw HTML, YouTube extracts links from the description, and PubMed parses the ReferenceList from official XML data.
- **Domain Authority (20%):** Verified institutional sites such as NCBI and universities receive the highest weight. Generic commercial or spam domains are scored conservatively.
- **Recency Score (20%):** A decay-based score that rewards content published within the last 12–24 months and penalizes content older than five years.
- **Medical Disclaimer (15%):** For health-related topics, the presence of a disclaimer such as "not medical advice" is rewarded to identify responsible publishing practices.

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4. Edge Case Handling & Robustness

- **Metadata Sanitization:** Noisy author strings such as CSS class artifacts are automatically cleaned. Inconsistent date formats are normalized to YYYY-MM-DD.
- **Transcript Failures:** YouTube videos without captions automatically fall back to the description, ensuring every source has a content body.
- **Link Density Penalty:** The system calculates the link-to-word ratio. Excessive link density triggers a score penalty to prevent link-farm blogs from achieving high trust scores.
- **Medical Bias Prevention:** PubMed sources receive a standardized +0.2 citation boost to reflect the peer-review process. Unverified health blogs are penalized when disclaimers are absent.

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5. Conclusion

This pipeline successfully combines robust multi-source extraction with NLP-based topic tagging and a nuanced trust scoring model. The resulting JSON outputs provide a clean, structured, and reliable dataset suitable for further analysis or downstream machine learning applications.